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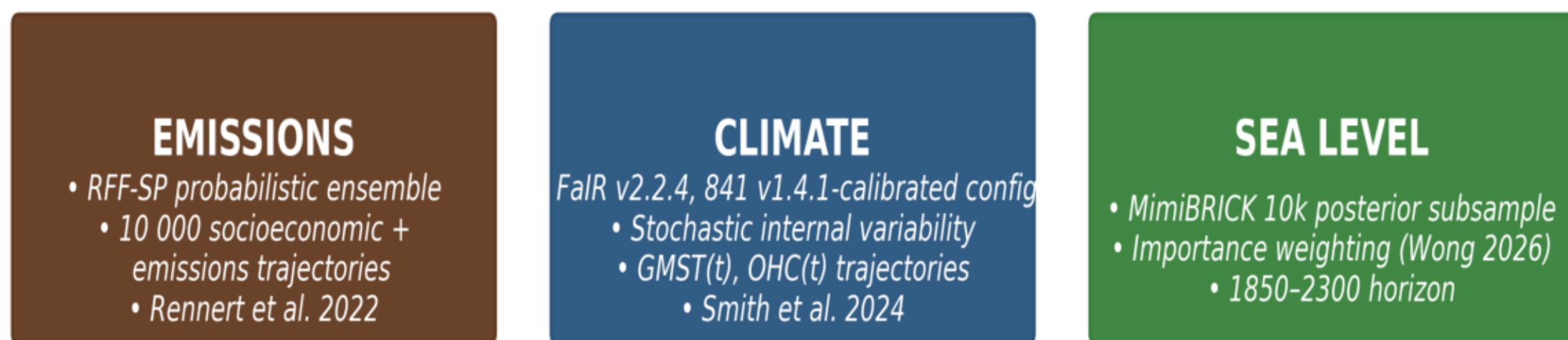
Simplified Damage Functions for Probabilistic Coastal Impact Estimation Using FrEDI

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Code: github.com/msarofim/SLR-RFF-BRICK • Data: doi.org/10.5281/zenodo.20451296

Read more: thesnapreport.substack.com

A. PIPELINE



DAMAGE-FUNCTION TRANSLATION (FrEDI)

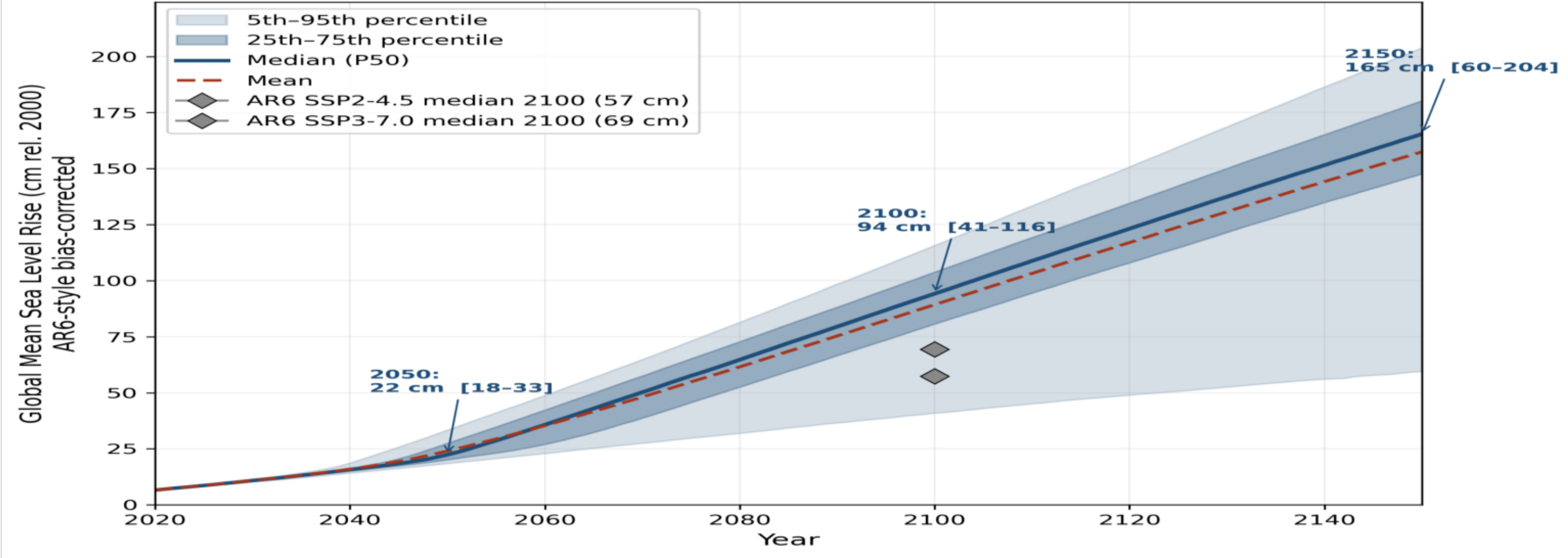
Sweet et al. (2022) NCA5 scenarios as interpolation nodes (0, 30, 50, 100, 150, 200, 250, 300 cm by 2100)
11-year rolling average to smooth capital-investment lumps • Linear interpolation between bracketing SLR-cm scenarios per year



→ STATE-LEVEL DAMAGES PER (YEAR, SCENARIO, ADAPTATION VARIANT) ←

METHODS: Each RFF-SP emissions trajectory drives a stochastic FaIR climate run; GMST(t) and OHC(t) feed MimiBRICK to give a probabilistic global SLR trajectory. BRICK draws are importance-weighted against observed GMSL (Wong 2026 AR(1) likelihood). Per-year global SLR is matched to bracketing Sweet (2022) NCA5 nodes; FrEDI interpolates linearly between bracket-specific damage functions and applies an 11-yr rolling average to smooth lumpy capital investments. Output: per-(year, sector, state, scenario, adaptation variant) damages in 2015 USD.

B. PROBABILISTIC SLR



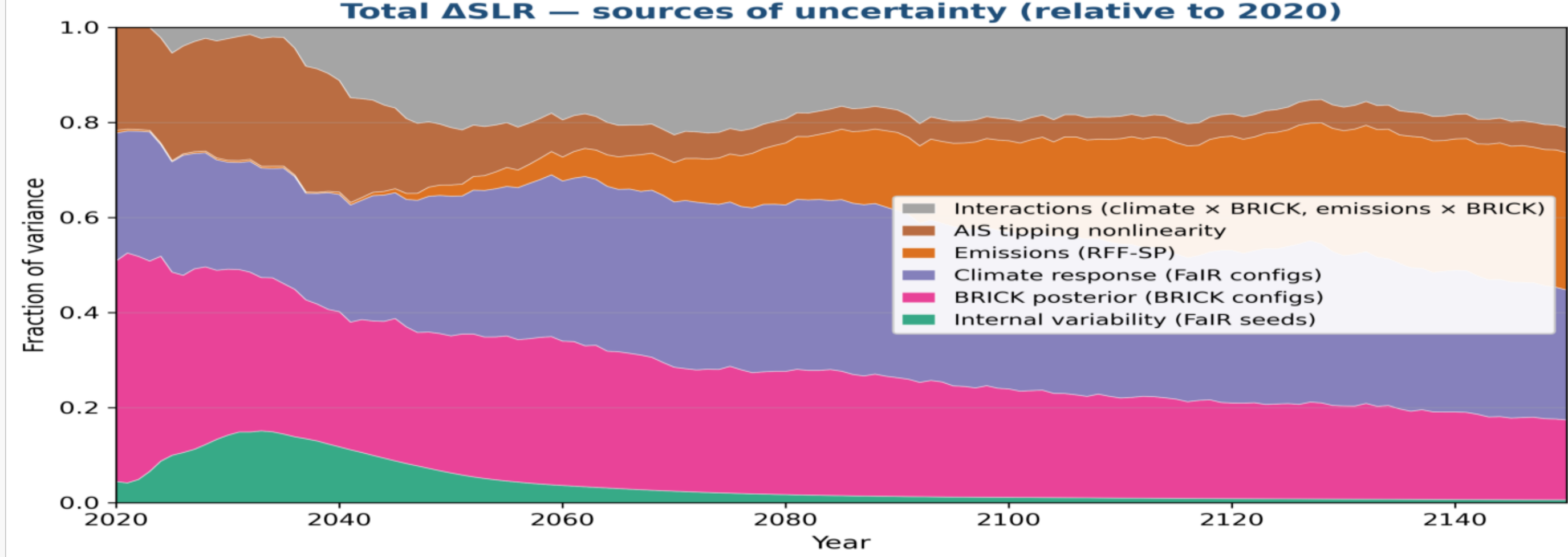
DISCUSSION

There is high value in probabilistic damage assessment for both mitigation-benefit estimates and adaptation-regional planning. Here we demonstrate one methodology that leverages emissions uncertainty from RFF, climate uncertainty from FaIR, sea-level-response uncertainty from BRICK, global-to-local mapping from Sweet et al. and damage functions from FrEDI to produce probabilistic damage estimates by US state for three SLR-sensitive impact sectors: coastal property damage, high-tide-flooding transportation interruptions, and high-tide-flooding elder mortality. This pipeline allows fast estimation of damages for any future scenario.

Louisiana is the state most at threat for SLR-related impacts, particularly on a per-capita basis. Florida, Massachusetts, Virginia, and New Jersey are distant 2nd-5th in per-capita impacts (Panel II). Among FrEDI sectors, HTF elder mortality has the largest monetized impact (Panel I), with HTF transportation 2nd and coastal properties 3rd (Panel II). Climate-response uncertainty is the largest contributor to total-SLR variance (≈33%) and marginal pulse response variance (≈44%) in 2100, with BRICK ice-sheet uncertainty (posterior + AIS tipping) second (≈28% / ≈41%); emissions uncertainty matters for total SLR (<10%) but is nearly irrelevant for the per-tonne marginal response (≈1%), which is essentially scenario-independent (Panels C, D).

Key considerations we identify for this kind of work: dependence between FrEDI and BRICK parameter uncertainty (Wong et al. 2026); look-ahead-based adaptation estimates that account for observed non-optimal adaptive behavior and smoothing of capital expenditures (Panel I); and the importance of a wide range of anchor scenarios for damage-function calibration (Panel F). The most important limitation we identify is the limited set of impact methodologies currently in FrEDI. We call on the community to produce more impact estimates that can be transformed into damage functions to inform this kind of analysis.

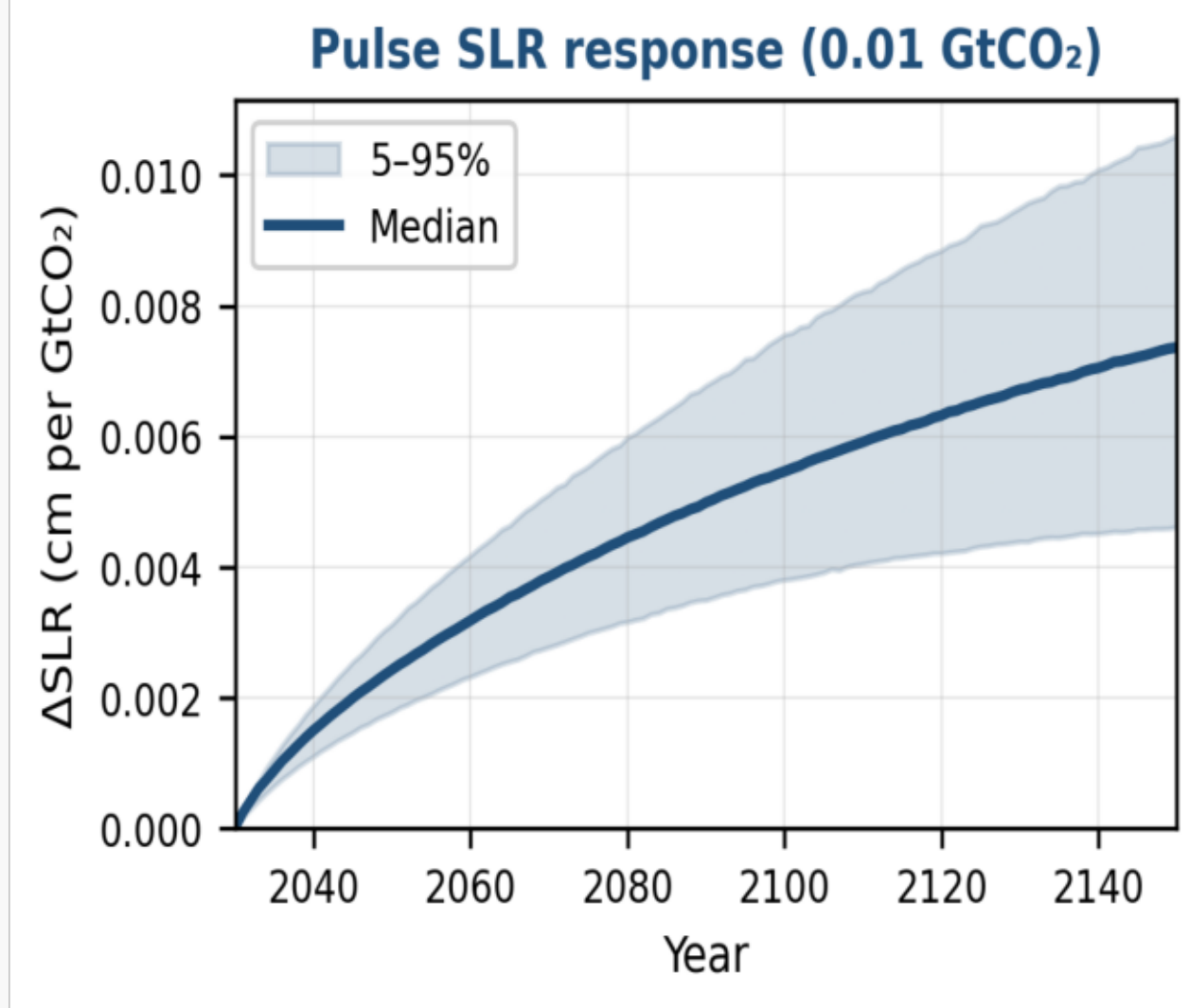
C. TOTAL SLR — sources of uncertainty



Heatmap: Scatter-plot decomposition of total-SLR variance, importance-weighted factorial design over 400 RFF-SP emissions paths, 51 climate calibrations per path, 17 FaIR scenarios seeds, and 25 BRICK posterior samples. See also: Global Mean Sea Level Rise: A Review of the State of the Science (IPCC AR6 WGII Chapter 10.2.2.2)

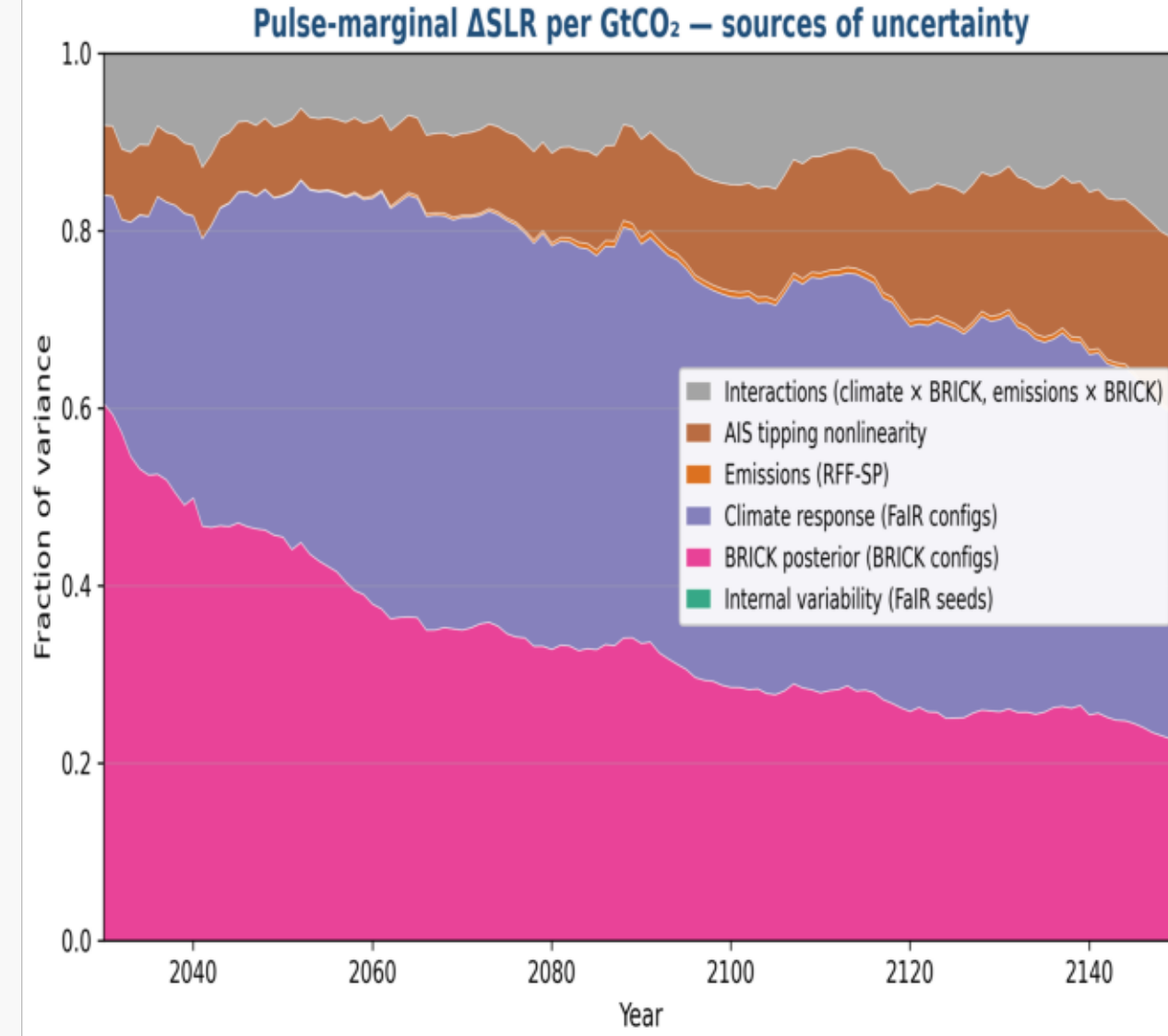
Conceptual companion to Daniel et al. 2025 (that Clim Change), who decompose total-SLR uncertainty across emissions vs. geographical sources at the multi-century horizon.

D. PULSE SLR RESPONSE



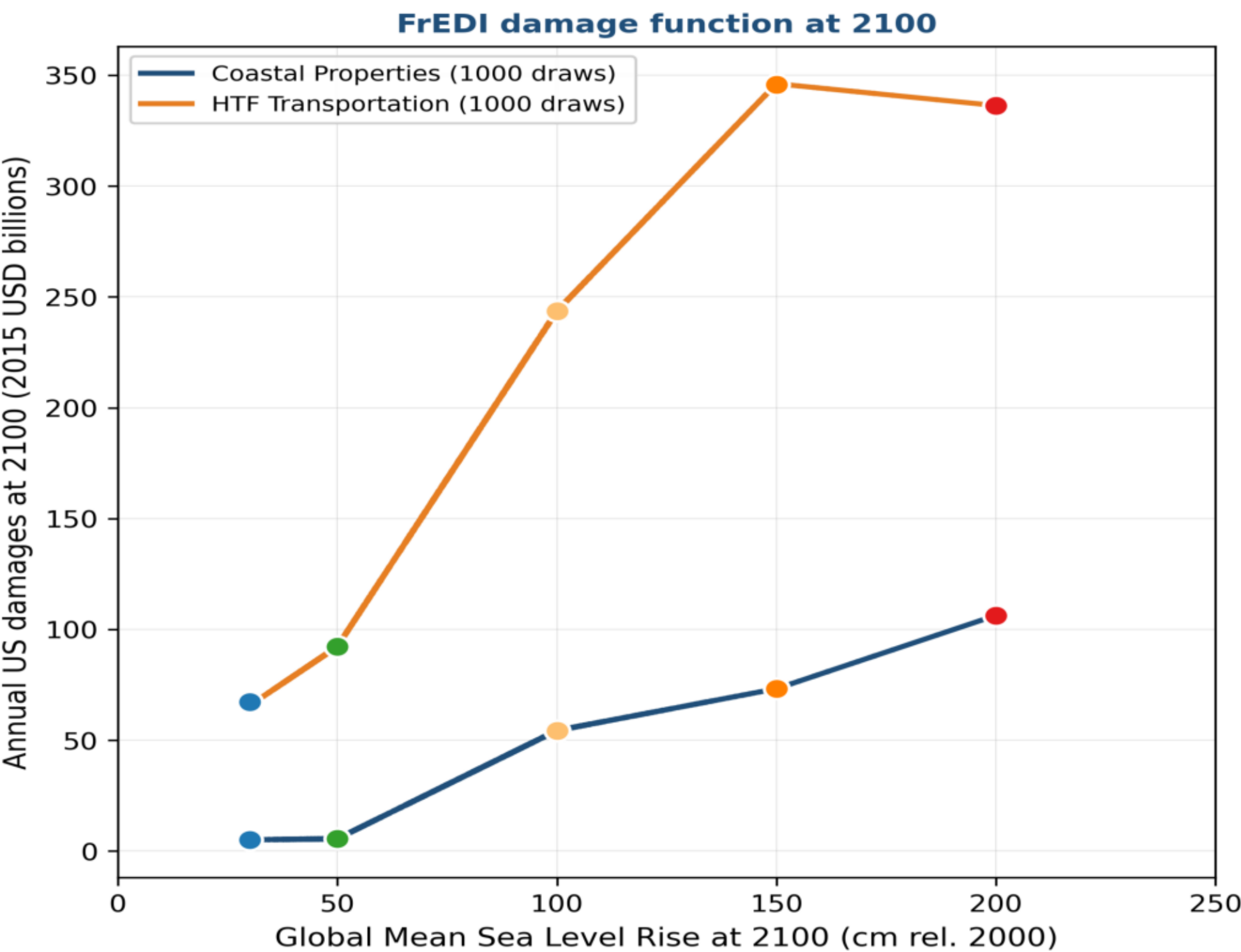
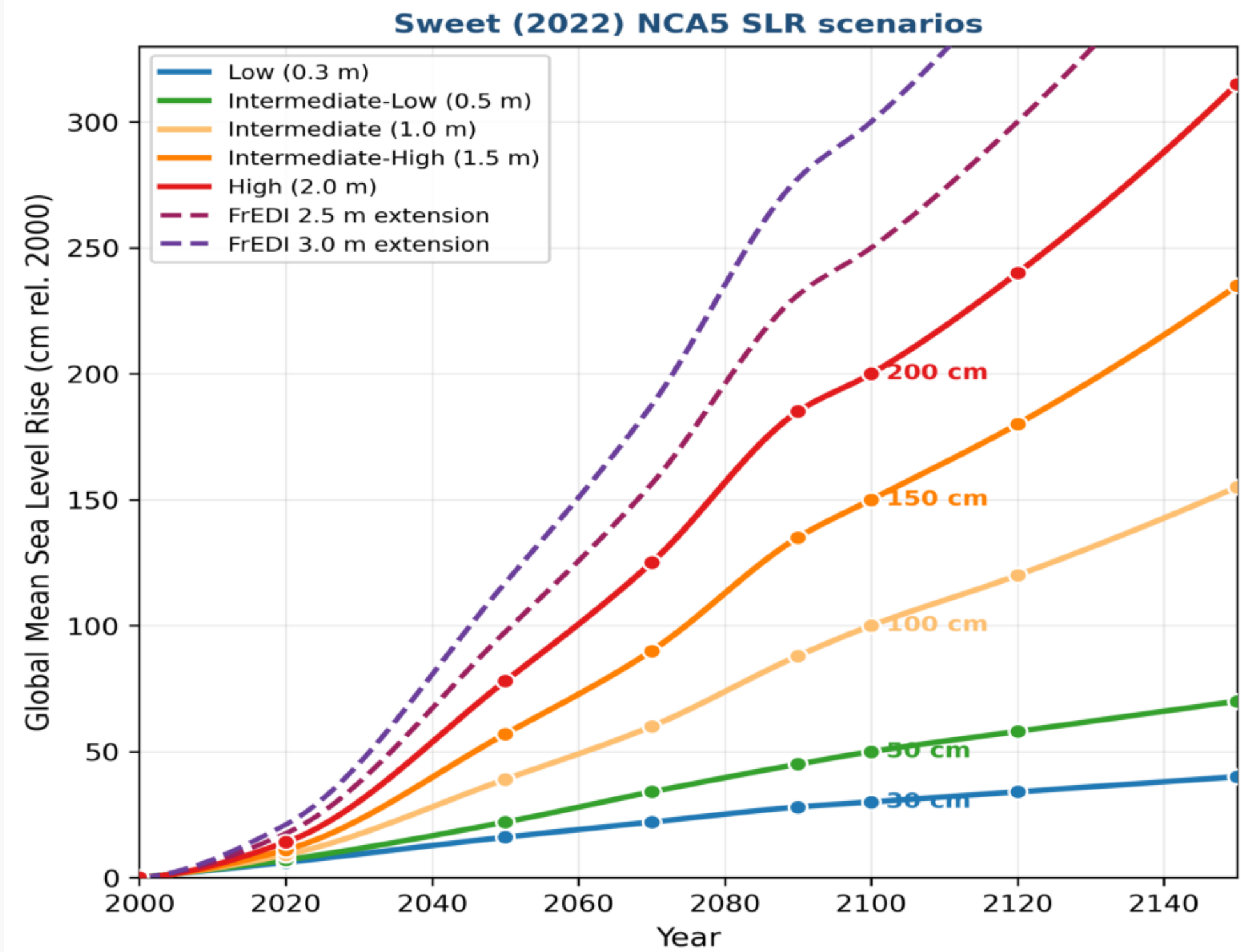
Total pulse-marginal SLR — median + 5-95% band, ASLR per GtCO2, from the 0.01-GtCO2 impulse arm (0.01-GtCO2-relevant three response, pulse-wise averaged) to illustrate how it compares to a new AIS-tipping draw.

D'. PULSE SLR — sources of uncertainty



Paired BRICK runs on the same factorial design as Panel C (12 GtCO2 pulse at 2030); color-coded fractions of the pulse-marginal ASLR.

F. DAMAGE-FUNCTION METHODOLOGY — Sweet 6 SLR nodes + FrEDI damage curve



Left: 6 NCA5 scenarios + 2 FrEDI extensions, response-cubic spline through every Sweet anchor year.
Right: empirical FrEDI damage function at 2100 from 1000 SLR-resampled RFF-SP draws (v1.4.3 v1.5-10k baseline); colored dots mark the Sweet calibration nodes that FrEDI interpolates between.

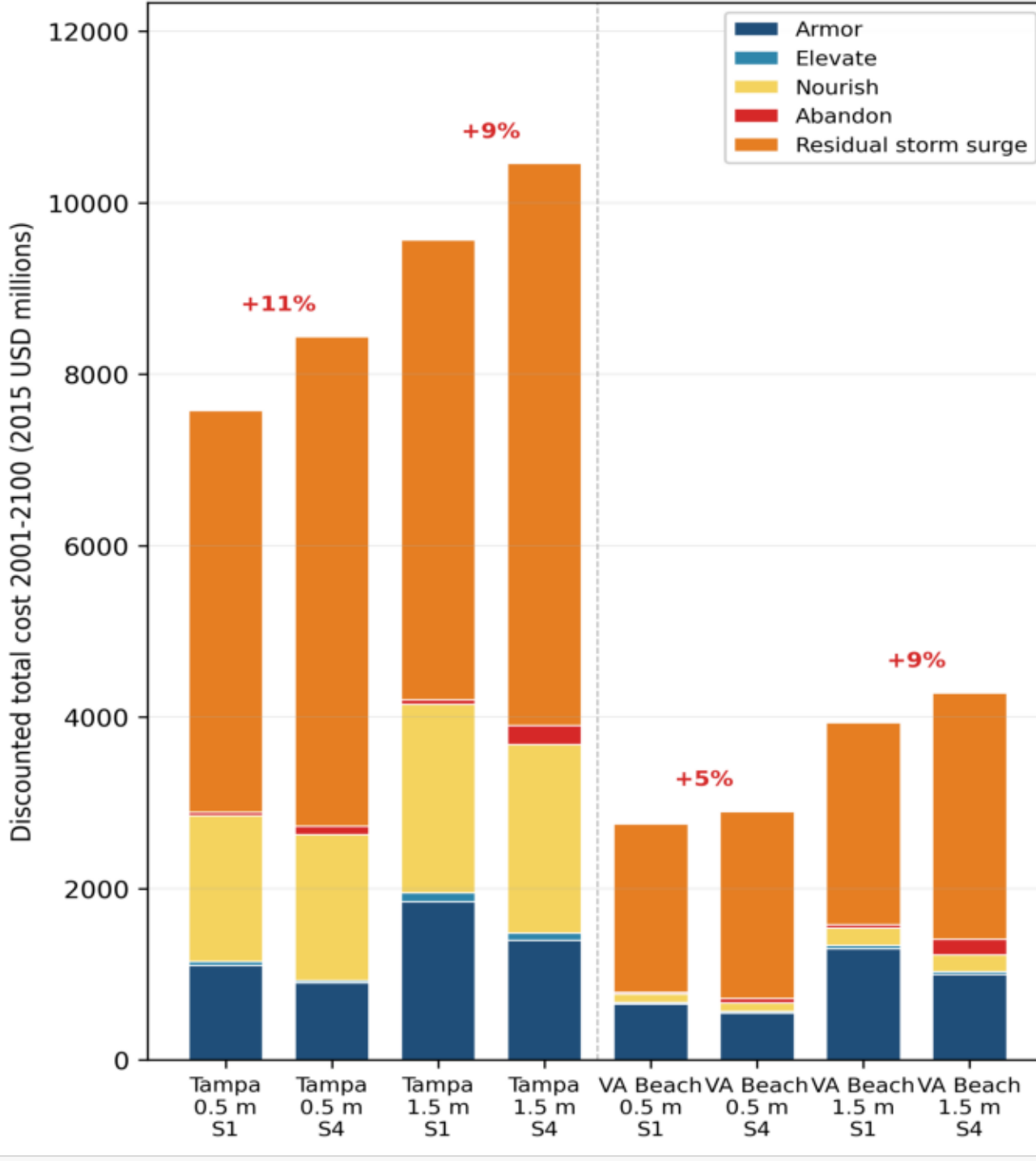
G. ADAPTATION (Lorie 2020) — NCPM decision logic + 11-yr smoothing

A. NCPM benefit-cost decision logic

(per 150-m grid cell, decade-by-decade — Lorie 2020 \$2.4.2)

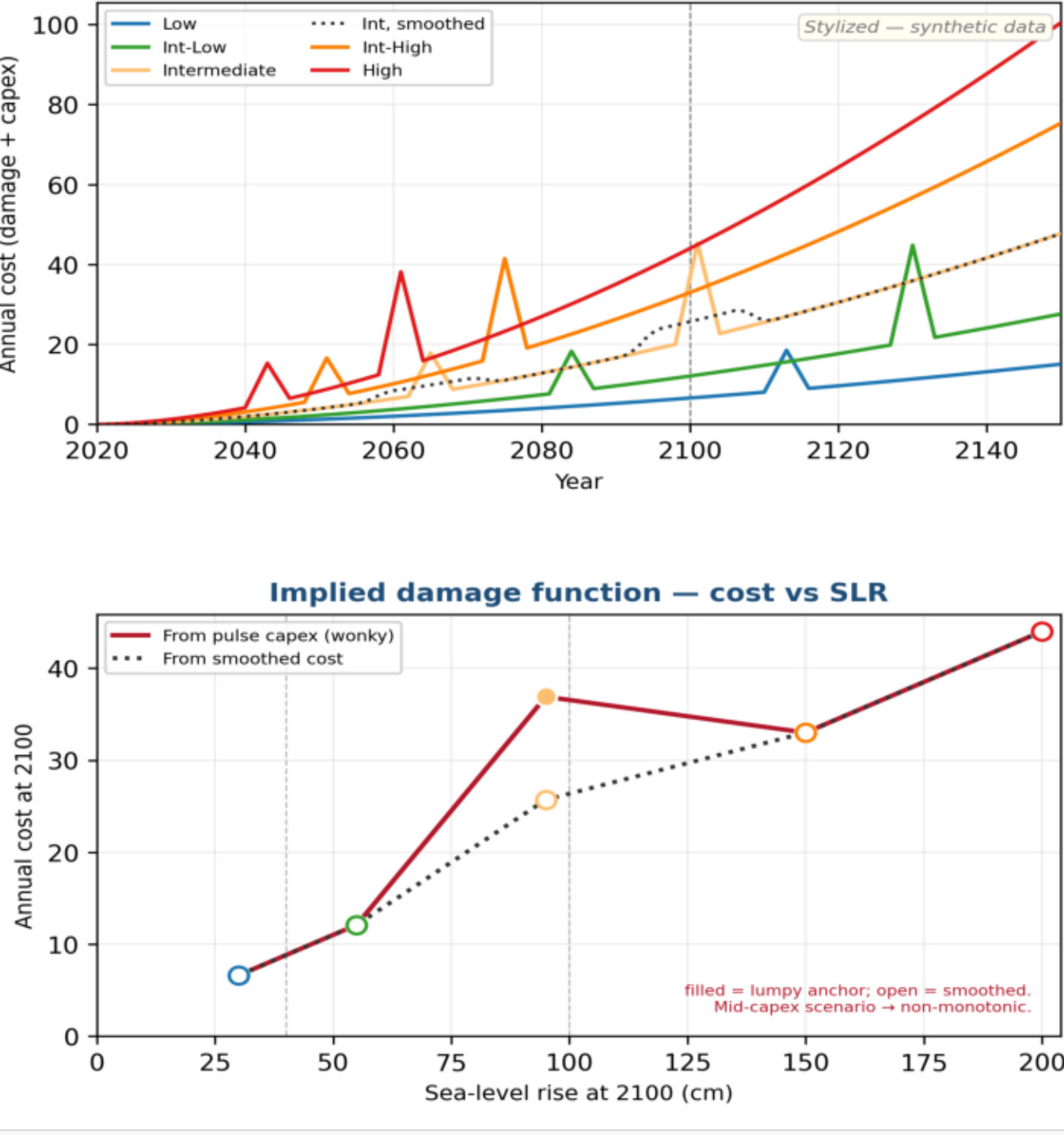
1. If cell not at risk from 100-yr storm surge over next decade → no action.
2. Else: compare 30-yr discounted avoided damages (EAD reduction) vs. cost of adaptation. If BCR ≥ 5 → armor or elevate. Tested at S=1 (optimal) and S=4 (sub-optimal).
3. Else: if 10-yr expected damages > property value → abandon. Otherwise no adaptation.

B. Adaptation + residual storm-surge costs

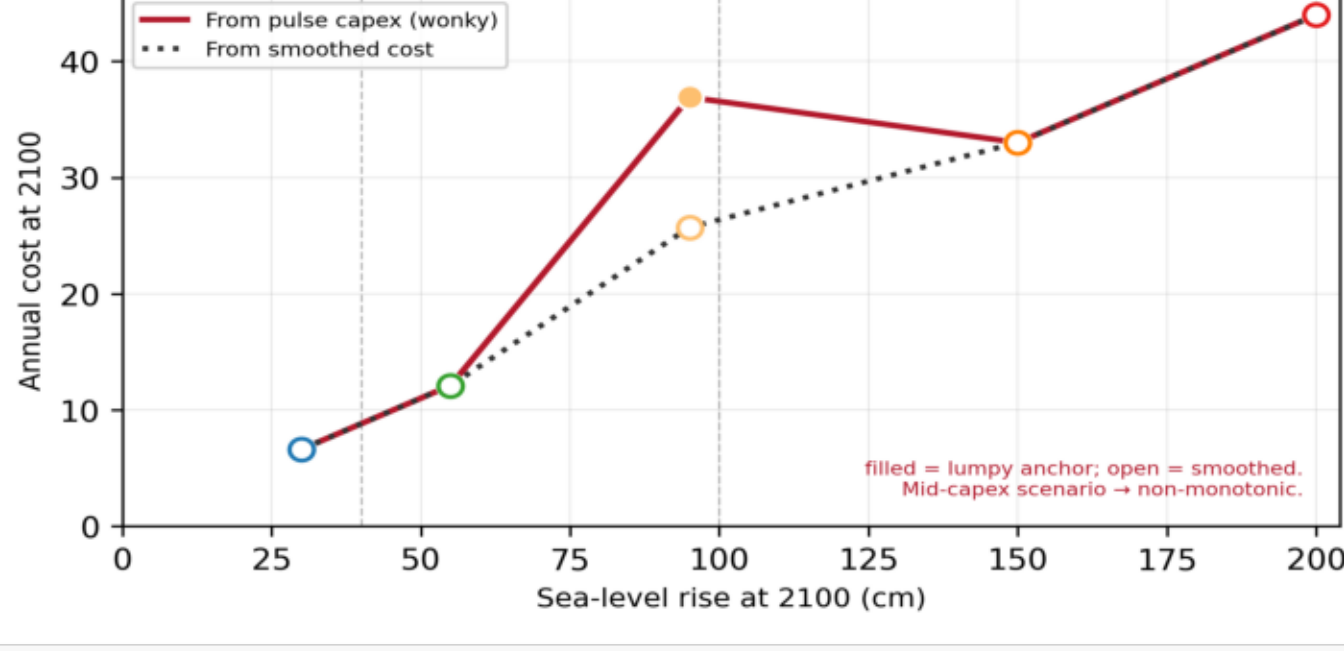


Left: NCPM decision logic with sub-optimal S=4 case representing observed under-adaptation (capital invested only when benefits exceed costs by ≥ 4×).
Middle: Lorie Table 2 cell shades for Tampa & Virginia Beach. Right: 21-year rolling-average smoothing of lumpy RFFW capital investments.
This smoothing is not only necessary because FrEDI's damage functions abstract away time — it is also consistent with how adaptation capital is amortized and frequently smoothed in the real world.

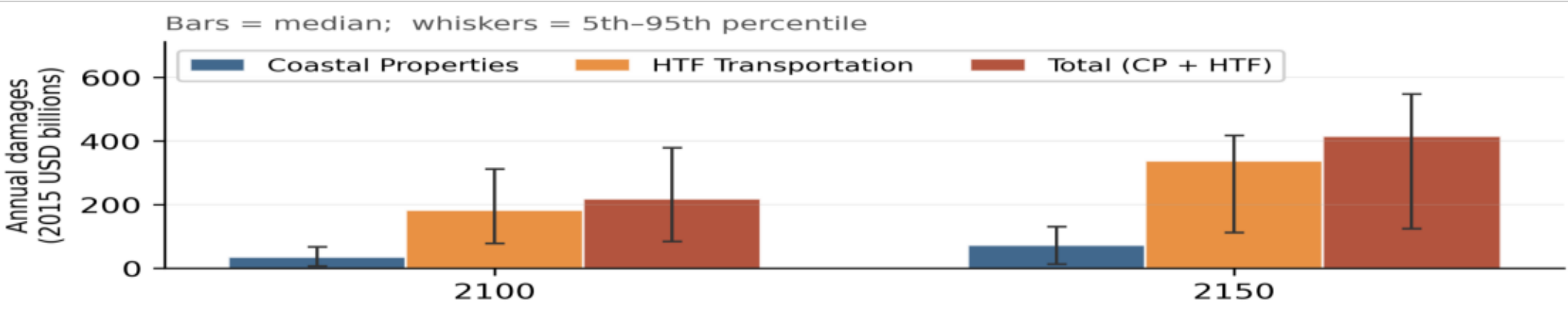
C. Lumpy adaptation capex → annual cost over time



Implied damage function — cost vs SLR



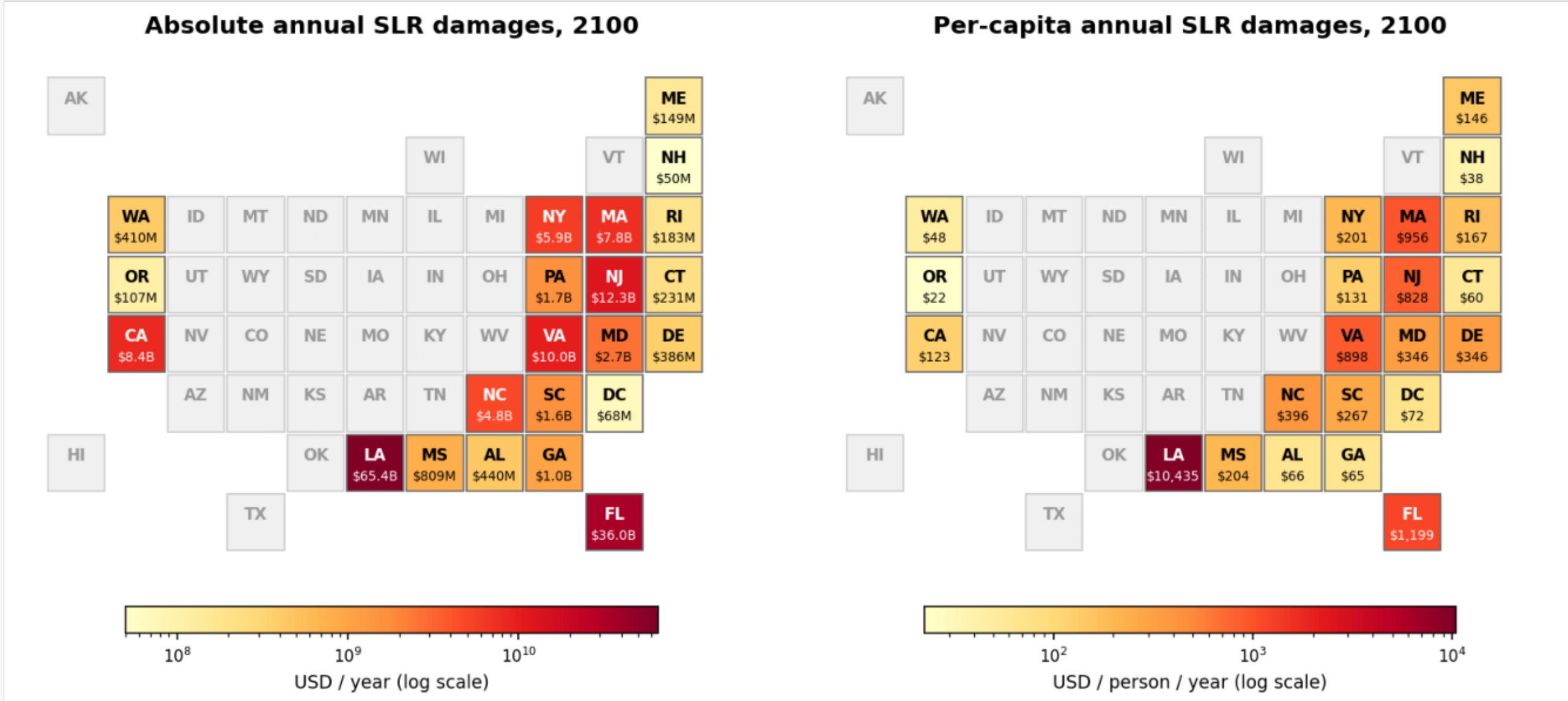
H. COASTAL PROPERTIES AND HTF TRANSPORTATION DAMAGES



	Year	P5	Median (P50)	P95	Mean
Coastal Properties	2100	\$5.3 B	\$35 B	\$67 B	\$34 B
	2150	\$12 B	\$73 B	\$130 B	\$73 B
HTF Transportation	2100	\$78 B	\$183 B	\$311 B	\$183 B
	2150	\$112 B	\$337 B	\$417 B	\$294 B
Total (CP + HTF)	2100	\$83 B	\$218 B	\$378 B	\$217 B
	2150	\$124 B	\$415 B	\$547 B	\$367 B

RFF-SP baseline ensemble (1,000 SLR-resampled draws from v1.4.3 v1.5-10k baseline; equal-weighted after SNI),
averages for Coastal Properties and HTF Transportation at 2100 and 2150.
Top (left) bars: median ± 5-95% whiskers.

I. COASTAL PROPERTIES AND HTF TRANSPORTATION DAMAGES BY STATE



Importance-weighted median annual damages (Coastal Properties + HTF Transportation) by state, 2100.
Absolute (USD) and per-capita (USD).

Louisiana is most impacted on a per-capita basis; FL/MA/VA are distant 2nd-5th.

J. HTF ELDER MORTALITY (Sheehan 2025)

	5th percentile	50th (median)	95th percentile
— No additional adaptation (Sheehan 2025 Table 2) —			
Additional deaths/yr, 2020	206	228	244
Additional deaths/yr, 2050	1,381	1,515	1,996
Additional deaths/yr, 2100	4,281	9,661	27,963
VSL-monetized damages, 2020 (USD billion)	\$2.8	\$3.1	\$3.4
VSL-monetized damages, 2050 (USD billion)	\$30.0	\$32.9	\$43.3
VSL-monetized damages, 2100 (USD billion)	\$170.9	\$385.6	\$1,116.0
— With stylized adaptation (cap at 95th-pct baseline) (Table 3) —			
Additional deaths/yr, 2100	2,765	5,555	13,312
VSL-monetized damages, 2100 (USD billion)	\$110.3	\$221.7	\$531.3
— Reduction at 2100 from stylized adaptation —	≈ 35%	≈ 43%	≈ 52%

Sheehan et al. 2025 Tables 2 & 3. Additional 65+ deaths / yr from high-tide flooding under the Rennert et al. (2022) RFF-SP ensemble (1000 draws) at 2100, v1.4.3 v1.5-10k baseline.

Bottom row: % reduction from stylized adaptation at the 5th, 50th, and 95th percentiles.

K. CAVEATS

- RFF-SP emissions vintage: Scenarios were developed about 5 years ago; given rapid renewable / battery advances, present-day analogs might differ.
- Fair known limitations: State-of-the-art probabilistic implementation, but missing permafrost feedbacks, possible Amazon dieback, and ozone / carbon-cycle interactions.
- BRICK upper bias: Runs slightly higher than the AR6 best estimates.
- Sweet et al. as a single realization: One realization of local SLR given global SLR; different partitioning between AIS, Greenland, thermal expansion, and local wind / current effects would yield different results.
- FrEDI sector coverage: Accounts for a limited number of SLR-derived damage categories; does not yet include probabilistic damages.

L. REFERENCES

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